

# The nontrivial contractively complemented subspaces of $H^p$

A later affirmative answer to the closing question in arXiv:math/0504294

Literature-status identification, lane 12

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**Status.** `literature_already_answered`. The later paper of Fu, Guo, and Li classifies all contractive projections and all nontrivial 1-complemented subspaces of  $H^p(\mathbb{T})$  for  $1 \leq p < \infty$ ,  $p \neq 2$ .

## Source question

Lancien, Randrianantoanina, and Ricard introduce, for an inner function  $\psi$  with  $\psi(0) = 0$  and a congruence class  $\Lambda = \{qk + r : k \in \mathbb{N}_0\}$ , the 1-complemented space

$$H_{p,\Lambda}^\psi = \overline{\text{span}}^{H^p} \{\psi^n : n \in \Lambda\}.$$

At the end of Section 4, printed p. 8, they ask:

“We do not know whether every 1-complemented subspace of  $H_p$  has to be isometric to  $H_{p,\Lambda}^\psi$ , for some coset  $\Lambda$  in  $\mathbb{N}$  and some inner function  $\psi$  vanishing at the origin.”

This follows a subsection explicitly devoted to infinite-dimensional ranges. The intended question is therefore about nontrivial (rank  $> 1$ ) ranges; one-dimensional ranges are the elementary exception already present in every Banach space [1].

## Decisive later theorem

Fu, Guo, and Li cite the 2005 paper and state that they will give explicit expressions for all contractive projections and 1-complemented subspaces of  $H^p$  [2]. Their Theorem 1.2 says that every contractive projection  $P : H^p \rightarrow H^p$  of rank greater than one is

$$P = \varphi \mathbb{E}_\varphi(\cdot / \varphi \mid \eta) \big|_{H^p}$$

for an admissible inner function  $\eta$  and multiplier  $\varphi$ , and its range is

$$H_{\eta,\varphi}^p = \{\varphi(f \circ \eta) : f \in H^p\}.$$

Definition 1.1 and Proposition 4.2 identify every such range isometrically with  $H^p$ . The paper records the consequence explicitly:

**Theorem 1** (Fu–Guo–Li, Corollary 1.4). *Let  $1 \leq p < \infty$ ,  $p \neq 2$ . Every nontrivial 1-complemented subspace of  $H^p$  is isometric to  $H^p$ .*

## Why this answers the 2005 formulation

The full Hardy space is one of the 2005 model spaces: take  $\psi(z) = z$  and the full congruence class  $\Lambda = \mathbb{N}_0$ . Then

$$H_{p,\Lambda}^\psi = H^p.$$

Consequently every nontrivial 1-complemented subspace is isometric to a space of the precise form proposed in the source. The answer is affirmative for all parameters in the source,  $1 \leq p < \infty$ ,  $p \neq 2$ .

The later classification is actually stronger: it describes the range as an embedded weighted-composition space  $H_{\eta,\varphi}^p$  and gives the projection itself, not merely its abstract isometry type.

## Evidence and scope

The identification used local full sources for arXiv:math/0504294 and arXiv:2412.14808v4, exact-phrase searches, and theorem-level inspection. The original and supporting PDFs are stored beside this note. No part of the  $p = 2$  Hilbert-space case is claimed; it was excluded in the source question.

## References

- [1] F. Lancien, B. Randrianantoanina, and E. Ricard, *On contractive projections in Hardy spaces*, Studia Math. 171 (2005), 93–102; arXiv:math/0504294.
- [2] X. Fu, K. Guo, and D. Li, *Extension of contractive projections*, arXiv:2412.14808v4 (2025), especially Definition 1.1, Theorem 1.2, and Corollary 1.4.